Research on Employee Performance Prediction Based on Machine Learning

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***Abstract*—This paper constructs a prediction model based on machine learning algorithm and enterprise human big data, and predicts the performance of enterprise employees. Empirical research shows that this method and the performance prediction model can accurately predict employee performance. At the same time, in the data construction stage, it focuses on the collection of employee attendance information and data construction. The experimental results show that this type of data is feasible and accurate for accurately predicting employee performance.**

Keywords—machine learning, performance, prediction

I. INTRODUCTION

Performance is the synthesis of achievements and benefits [1]. It is the synthesis of the thoughts, behaviors, methods, results and objective effects of an organization or individual in a certain period of time. In short, performance is the synthesis of input and output of an organization or individual in a certain period of time. Input refers to material resources such as human, material and time, and output refers to the achievement of a certain goal in terms of quantity and efficiency. From the perspective of enterprise management, performance is the various outputs of enterprises at different levels in order to achieve their goals. It is the expectation and purpose of enterprise management [2]. At this time, performance is divided into enterprise performance and individual performance. The realization of enterprise performance is based on and depends on individual performance. If the enterprise performance can be decomposed to each employee according to a certain logical relationship, as long as each employee can better meet the enterprise requirements, the enterprise performance can be better realized [3].

However, employee performance is often affected by many factors, such as individuals, skills, incentives, opportunities, environment and so on. The specific factors and details of each aspect may have a great impact on employee performance. For enterprise management, controlling these factors is equal to controlling individual performance. At the same time, if we can predict the performance output according to some of these factors, it is possible to strengthen the control of these factors affecting performance, so as to improve personal performance. Therefore, whether an enterprise can accurately predict and judge employee performance is directly related to the stable growth of enterprise organization and performance, which is of great significance to enterprise management. Therefore, the research on enterprise employee performance prediction has always been a hot spot in the field of enterprise management [4-7].

In the past, performance prediction is usually expected to start with the factors that directly affect employee performance. These direct factors are divided into two types: employees' internal characteristics and external environment [8]. Internal characteristics such as personality, ability and their determinants, and external environment such as tools and processes and their determinants. However, for a long time, the data collection and construction of these direct factors and their determinants often affect the accuracy and practicability of performance prediction because of practical difficulties, which has become a relevant research difficulty. In fact, in enterprise production management, some information and data are produced for a long time and easy to collect. These information and data seem not to be directly related to employee performance, but it is possible to predict employee performance because of some hidden connection, such as communication, attendance information, etc.

Attendance information is a common basis for enterprises to assess employees' performance. It directly reflects employees' compliance with corporate disciplines, regulations and other binding requirements, and indirectly reflects employees' subjective satisfaction and objective conflict relations in all aspects of the enterprise, such as job satisfaction, environmental satisfaction, relationship satisfaction, job engagement (emotional commitment), workload, awareness of promotion opportunities, etc. Therefore, attendance information is an objective data combined with employees' subjective will, which also determines the richness and complexity of the implicit information of this type of data. Enterprise attendance information comes from various time records of employees' daily work (such as off-duty time, overtime, etc.) or abnormal situation records (such as business trip, official business, vacation, sick leave, maternity leave, etc.) by the human resources management department, which belongs to human resources data. The human resources department of an enterprise usually counts the attendance information of employees in each department on a monthly basis and completes relevant processing and storage. The data after attendance information processing usually includes monthly late days, monthly early leave days, monthly absenteeism days, monthly leave days (illness, work, maternity, leave, etc.), monthly off work or field days, etc.

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Due to the characteristics of easy collection, periodic generation, large scale, indirect correlation with prediction objectives and rich implicit information, if we can establish a certain relationship between it and performance, we may realize the prediction of performance based on attendance information.

Big data mining is a process of automatically extracting useful information hidden in data from large data sets combined with machine learning methods. These useful information can be expressed as laws, rules, relationships and patterns, which can provide support for decision-making. The technology to extract hidden information or establish relationships is big data mining technology. It can be seen that big data mining technology is a sharp tool to "crack" the hidden information in the data. If it is combined with attendance data, it can "mine" a sufficiently accurate relationship model between employee performance and attendance information, and then the attendance information can be applied to predict employee performance.

Up to now, the research on applying big data mining method to enterprise employee performance prediction is relatively rare. In this paper, we predict employee performance and verify its feasibility with the help of big data mining technology and the massive basic data - attendance information that can be provided for a long time in enterprise operation and management activities. Firstly, a general construction scheme of attendance data is proposed, and employee data sets are collected and constructed based on this method. Then, based on the data set, the performance prediction model is constructed by using different machine learning algorithms under the unified data framework. Then, by comparing the performance of different performance prediction models, select the best employee performance prediction model and predict employee performance. Empirical research shows that this method can accurately predict employee performance and provide strong support for enterprise human resource management decision-making.

II. Data Mining Method of Employee Performance Prediction

A. Thinking Framework

Data mining technology, such as regression tree, SVM, neural network and so on, has been widely used in many fields, but at present, there is less research on enterprise employee performance prediction combined with machine learning method. This paper provides a machine learning method based on big data mining to predict enterprise employee performance, and focuses on the feasibility of introducing employee attendance data to predict employee performance.

The logical framework of employee performance prediction research based on machine learning is mainly divided into five parts, namely data collection, employee data set construction, prediction model establishment based on machine learning, selection of optimal prediction model and employee performance prediction. Data collection and employee data set construction is to provide employee original data set for data mining modeling. Employee data set usually needs to be further divided into training set and test set for subsequent machine learning modeling and evaluation of optimal prediction model. Model building is the process of training data through a variety of machine learning algorithms and establishing corresponding performance prediction models. The performance prediction model is substituted into the "optimal" prediction model, and its performance needs to be tested. Employee performance prediction refers to the process of predicting the future performance of employees by using the selected optimal prediction model and combined with the "new" employee characteristic data, so as to provide a basis for human resource management decision-making for a long time.

B. Data Set Construction

The data set of enterprise employee performance prediction research can be completed through three steps: data collection, data sorting and feature selection.

Focus on collecting basic data such as employee identity information, performance information and attendance information through the administrative department, human resources department and other departments of the enterprise, especially the attendance data should be collected as fully and accurately as possible. These data are also called original data:

1) identity information: it includes information about employee name, age, registered residence, address, marital status, education, major, technical ability, working years, department, work type, rank, training experience, reward and punishment record, etc.

2) Performance information: it mainly includes the assessment information of employees' work performance, ability, attitude and morality by month or year, such as task completion rate, performance score, performance bonus, merit rating and other information;

3) Attendance information: it mainly includes the normal shift or overtime time, travel time, field time and other information counted by employees on a daily basis, as well as the time of illness, rest, maternity, personal leave and other information counted by month.

C. Modeling and Prediction Method

Modeling is model construction, which mainly includes two steps: model establishment and model selection. Prediction is to use the best prediction model to predict employee performance combined with new employee data.

The prediction of enterprise employee performance is actually to use employee data set to solve the regression prediction problem of employee big data mining. Regression models are often used to predict unknown values. The modeling and prediction of regression problem is to establish a regression model about the influence or decisive relationship of the predictive variable (dependent variable) with continuous value by using other variables (independent variables) of the sample through the analysis of the training set with known value, so as to use the regression model to predict the value of new test data (new sample). For example, to predict employee performance is to analyze the training set whose dependent variable value of employee performance is clearly known, so

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as to find the functional relationship between other independent variables and dependent variables in a training set. This functional relationship establishes a sufficiently accurate regression model that can use other independent variables to determine the value of dependent variables. Using this model, we can apply other employee data to estimate, fit or predict the employee's performance value.

III. Employee Performance Prediction Based on Machine Learning

A. Method Overview

The essence of machine learning method based on big data for enterprise employee performance prediction is to apply machine learning to solve the regression problem of big data mining, that is, use machine learning method to mine the massive data of enterprise employees, then build the best prediction model and predict the value of employee performance, so as to provide basis for enterprise human resource management decision-making.

B. Data Analysis

1) Data set construction

The data of this study comes from the human resources data of an information technology manufacturing company in Chengdu. The basic information data such as identity, attendance and performance of employees in recent three years are collected through the administrative management and human resources departments of the company. Through the data sorting of the original information, we can get the original data table that preliminarily reflects the employee's identity, attendance and performance. The main data, i.e. attendance and performance information variables, comes from the processing of information aggregation and deletion of daily attendance records and performance records of the company's employees for three consecutive years: the attendance records are first processed by month, including daily average attendance hours, daily average overtime hours, monthly absenteeism days, monthly lateness times, monthly average lateness hours (minutes), monthly early leave times, monthly average early leave time (minutes), monthly leave days, average daily clock in times and other attendance indicators, and then take the three-year average to get the final attendance information, that is, three-year average daily attendance time (hours), three-year average daily overtime time (hours), proportion of average absenteeism days in three years (proportion of monthly absenteeism days = monthly absenteeism days / monthly attendance days, in which monthly attendance days = monthly working days - field or business trip days - various personal leave days), average number of lateness in three years, average length of lateness in three years (minutes), average number of early leave in three years, average length of early leave in three years (minutes), average number of leave in three years attendance data such as average daily clocking times in three years; For the performance scoring data, take the average of three years to get the average annual performance of three years. At the same time, through the records of promotion, reward and training, get the performance data such as whether to be promoted, the number of annual excellent awards and the number of annual training.

Further, the Pearson correlation coefficient is used to calculate the correlation between various types of variables in the original employee data table and the dependent variables, and the variables with the highest correlation ranking are retained to recombine the data table with the dependent variables. In this way, the data table after data processing is the initial data set. The brief introduction of the data set is as follows:

There are 15 variables and 959 observation samples in the employee data set of performance prediction, which represent the basic data of 959 employees of an enterprise.

2) Data exploratory analysis

There are 15 independent variables in the employee data set, belonging to three basic types:

Identity information variables: gender, age, marital status, educational background, etc.;

Occupational information variables: Department, rank, seniority, etc.;

Attendance information variables: three-year average daily attendance hours, three-year average daily overtime hours, the proportion of three-year average absenteeism days, three-year average late times, three-year average late hours, three-year average early leave times, three-year average early leave hours, etc.

Firstly, exploratory analysis of employee data.

Based on the exploratory analysis of the above employee data, we can summarize the company's employee performance profile and corresponding suggestions:

- The company's performance evaluation system is more beneficial to older female employees with older age and length of service, higher rank and married;
- The average performance of employees with bachelor's degree is the lowest, indicating that there is still room for improvement in the performance of employees with this degree level;
- At present, the comparison of average performance between departments is balanced;
- The average daily overtime of employees is about half an hour, and the relationship between overtime and average performance is not obvious;
- The median daily attendance length of employees is 9 hours, which meets the company's requirements for punch in and attendance from 8:30 to 17:30, and this feature is positively correlated with the average performance;
- The median monthly absenteeism of employees is 9% and the average is 16%, indicating that the absenteeism of employees is serious. The eigenvalue is negatively correlated with average performance.

The above shows that the company's performance evaluation system is relatively balanced, but it lacks incentives

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for young employees. In addition, absenteeism is serious and needs attention.

C. Establishment and Selection of Prediction Model

1) Machine learning algorithm modeling

This section introduces the principles of five commonly used machine learning algorithm models and R language implementation methods. At the same time, it constructs prediction models combined with employee data sets, gives performance evaluation, and demonstrates the implementation process of prediction models of different machine learning algorithms in the form of examples.

- Linear regression

70% of the 959 samples in the employee data set are randomly selected as the training set and the remaining 30% as the test set. The number of samples in the training set and test set is shown in Table I. With the help of R language data mining tool, the training set and test set are used for ordinary least squares regression, robust regression, ridge regression model modeling and model performance prediction performance evaluation.

TABLE I. SAMPLE QUANTITY OF TRAINING SET AND TEST SET

Number of training sets	Number of inspection sets
673	286

The comparison of performance prediction effect indicators of three linear regression models is shown in Table II:

TABLE II. SAMPLE QUANTITY OF TRAINING SET AND TEST SET COMPARISON OF PREDICTION PERFORMANCE INDEXES OF VARIOUS LINEAR REGRESSION MODELS

Model	RMSE	R2
Ordinary least squares regression	5.922	0.426
Robust regression	5.936	0.423
Ridge regression	5.922	0.426

Comparing the prediction indexes of three common linear regression models, from the prediction error effect RMSE index, the prediction effect of ordinary least squares regression is the same as that of ridge regression, but both are better than robust regression; From the R2 index representing the prediction accuracy of the model, the accuracy of ordinary least squares regression is the same as that of ridge regression, and both are better than robust regression. At the same time, it can be seen that the difference between the two indexes of the three linear regression models is very small, so it can be considered that the prediction effects of the three linear models are similar.

- Nonlinear regression

70% of the 959 samples from the employee data set are randomly selected as the training set and the remaining 30% as the test set, so the sample size of the training set is 673 and the sample size of the test set is 286. Select GLM models corresponding to three common combinations of distribution and connection functions, namely Gaussian (link = "identity"), Poisson (link = "identity") and gamma (link = "identity"), for modeling performance prediction and performance evaluation.

The scatter diagram of performance prediction error of three GLM models is shown in Figure 1:

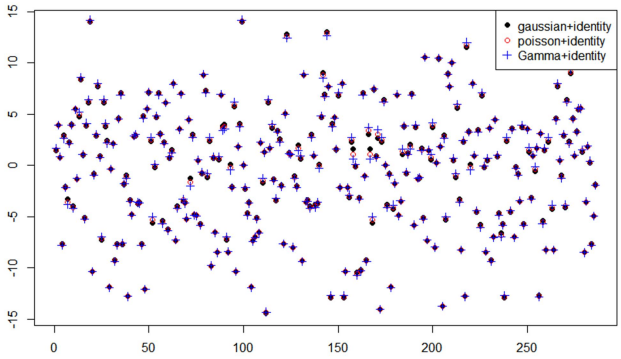


Fig. 1. Scatter diagram of model prediction error.

The predicted performance indicators of the corresponding model are shown in Table III.

TABLE III. COMPARISON OF MODEL PREDICTION PERFORMANCE INDICATORS

Model	Distribution and connection function	RMSE	R2
generalized regression	gaussian(link="identity")	5.922	0.426
	poisson(link="identity")	5.930	0.425
	Gamma(link="identity")	5.941	0.423

Comparing the prediction indexes of three commonly used linear regression models, combined with the prediction error RMSE index and the prediction accuracy R2 index, when the distribution function of GLM model takes Gaussian distribution and Poisson distribution, the effect is roughly the same, which is better than that of GLM distribution function taking gamma distribution. At the same time, it can still be seen that the numerical difference of the evaluation indexes of the three nonlinear regression models is very small, and it can be considered that the prediction effects of the three nonlinear models are similar. It should be noted that the first GLM model is equivalent to an ordinary linear model. Therefore, from the perspective of evaluation indicators, the performance prediction effect of the first GLM model is the same as that of the above-mentioned ordinary least squares linear model.

- Regression tree

70% of the 959 samples from the employee data set are randomly selected as the training set in proportion, and the rest are used as the test set, so the sample size of the training set is 673 and the sample size of the test set is 286.The training set and test set are used for regression tree model modeling and prediction performance evaluation.

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For regression tree modeling, "ANOVA" is selected as the continuous data type according to the dependent variable. The relationship between model cross validation error and complexity parameters is shown in Figure 2:

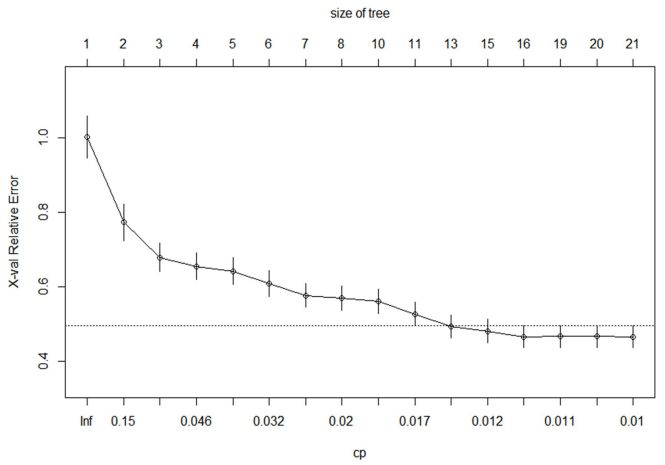


Fig. 2. Relationship between model cross validation error and CP.

The prediction performance evaluation of the regression tree model is given by RMSE and R2 indicators, as shown in Table IV:

TABLE IV. COMPARISON OF MODEL PREDICTION PERFORMANCE INDICATORS

Model	RMSE	R2
Regression tree	5.526	0.500

It can be seen that compared with the previous linear or nonlinear regression models RMSE and R2 indicators, the prediction performance of employee performance of regression tree is better than that of linear or nonlinear regression model.

- Support vector machine

70% of the 959 samples from the employee data set are randomly selected as the training set in proportion, and the rest are used as the test set, so the sample size of the training set is 673 and the sample size of the test set is 286.The training set and test set are used for SVM model modeling and prediction performance evaluation.

Support vector machine model can be used as classification model, regression model or anomaly detection model. The type parameter of SVM function is used to specify the model type of SVM. There are five optional types: c-classification, nu classification, one classification, EPS expression and nu expression, of which the first three are for classification model. The latter two are the classification models for numerical outcome variables, namely regression models. There are four optional functions for kernel parameters, among which "radial" and "polynomial" kernel functions have the best performance and are most commonly used. In addition, cost penalty function and gamma kernel parameters are important parameters of SVM.

The different values of SVM model category, kernel function, cost and gamma parameters have a great impact on the accuracy of the model. Therefore, it is necessary to use the exhaustive method to select the optimal model parameters for the regression prediction problem to obtain an accurate regression prediction model.

TABLE V. COMPARISON OF PREDICTION PERFORMANCE INDEXES OF DIFFERENT TYPES OF SVM MODELS

Model type	kernel function	Optimized cost parameters	Optimized gamma parameters	RMSE	R2
eps-reg	linear	100	0.1	5.972	0.416
eps-reg	polynomial	100	0.01	5.518	0.502
eps-reg	**radial**	**10**	**0.1**	**2.982**	**0.855**
eps-reg	sigmoid	100	0.001	5.961	0.419
nu-reg	linear	100	0.1	5.938	0.423
nu-reg	polynomial	1	0.1	4.686	0.641
nu-reg	radial	10	0.1	3.070	0.846
nu-reg	sigmoid	100	0.001	5.936	0.423

The comparison of prediction performance of SVM model under different parameter combination conditions is shown in Table V. It can be seen that the best model parameters are EPS reg type, radial kernel function, cost = 10 and gamma = 0.1.

As can be seen from the scatter diagram, compared with the prediction results of the above linear, nonlinear and regression tree models, the predicted value of SVM model is in good agreement with the true value. At the same time, the prediction error converges near 0, and the prediction performance is better. The prediction performance RMSE and R2 indexes of the SVM model are shown in Table VI. It can also be seen that compared with the above three types of models, the predicted RMSE index of this model has decreased significantly below 3, and the R2 index is close to 1, indicating that the prediction performance of employee performance has been ideal.

TABLE VI. COMPARISON OF MODEL PREDICTION PERFORMANCE INDICATORS

Model	RMSE	R2
SVM	2.982	0.855

- Random forest

The prediction performance RMSE and R2 indexes of the random forest model are shown in Table VII:

TABLE VII. COMPARISON OF MODEL PREDICTION PERFORMANCE INDICATORS

Model	RMSE	R2
Random forest	2.995	0.853

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It can be seen that, similar to the performance of SVM model, the RMSE index of the random forest model is below 3, and the R2 index is 0.853, close to 1. The performance of employee performance prediction has been ideal.

- Neural network

The prediction performance RMSE and R2 indexes of the neural network model are shown in Table VIII:

TABLE VIII. COMPARISON OF MODEL PREDICTION PERFORMANCE INDICATORS

Model	RMSE	R2
neural network	2.910	0.861

It can be seen that, similar to the performance of SVM and random forest model, the RMSE index of the neural network model is below 3, while the R2 index is 0.861, close to 1, and the employee performance prediction performance is also ideal.

2) Optimal prediction model selection

Based on the above design, modeling and comparative experiments are carried out. The R2 and RMSE of each prediction model are used as evaluation indexes to compare the prediction performance of each model. The effect evaluation is dominated by R2 index and supplemented by RMSE index. The comparison of R2 and RMSE indexes of each model is shown in Table IX and Table X respectively:

TABLE IX. COMPARISON OF AVERAGE R2 INDICATORS OF EACH MODEL

Model	Model type	Experiment 1	Experiment 2	Experiment 3	Experiment 4	Experiment 5	average value
OLS	/	0.445	0.423	0.48	0.517	0.349	0.443
Robustness	/	0.439	0.421	0.475	0.516	0.352	0.441
Ridge	/	0.445	0.422	0.48	0.517	0.349	0.443
GLM	gaussian	0.445	0.422	0.48	0.517	0.349	0.443
	poisson	0.445	0.422	0.481	0.515	0.348	0.442
	gamma	0.445	0.422	0.481	0.513	0.349	0.442
DT	/	0.553	0.505	0.558	0.569	0.472	0.531
SVM	eps-regression、radial	0.947	0.81	0.888	0.883	0.916	0.889
RF	/	**0.935**	0.881	0.893	0.891	**0.932**	**0.906**
NN	/	0.892	0.797	0.807	0.802	0.841	0.828

TABLE X. COMPARISON OF AVERAGE RMSE INDICATORS OF EACH MODEL

Model	Model type	Experiment 1	Experiment 2	Experiment 3	Experiment 4	Experiment 5	average value
OLS	/	5.819	5.612	5.893	5.478	5.765	5.713
Robustness	/	5.846	5.616	5.921	5.418	5.752	5.711
Ridge	/	5.819	5.612	5.893	5.478	5.765	5.713
GLM	gaussian	5.818	5.612	5.893	5.478	5.765	5.713
	poisson	5.817	5.608	5.889	5.489	5.766	5.714
	gamma	5.816	5.608	5.889	5.502	5.766	5.716
DT	/	5.222	5.193	5.436	5.176	5.195	5.244
SVM	eps-regression、radial	1.806	3.217	2.735	2.698	2.066	2.504
RF	/	1.989	2.548	2.673	2.606	**1.86**	**2.335**
NN	/	2.569	3.323	3.591	3.503	2.851	3.167

It can be seen from Table IX that the average R2 performance of the three linear regression models is basically consistent based on the average R2 index; The average R2 performance of the three nonlinear regression models is also basically the same; Among the six performance prediction models of linear regression, nonlinear regression, decision tree, SVM, random forest and neural network, the average R2 of random forest model is 0.906, the performance of SVM model is 0.889, and the average R2 of neural network model is 0.828. The performance of the above three models is significantly better than that of other models. The average R2 index performance of linear or nonlinear regression model is lower than 0.5, and the prediction performance is poor. From the stability of experimental results, among the six types of models, the R2 of each experiment of random forest model is higher than that of other models, and the index stability is the best. At the same time, from the performance of a single experiment, the optimal prediction comes from random forest, and the optimal R2 is as high as 0.935. Therefore, in general, the prediction performance of random forest model is the best, SVM is the second, and the performance of linear or nonlinear regression model is the worst.

Taking RMSE as the evaluation index, we can get basically similar conclusions from the experimental results in table X, that is, the average RMSE performance of random forest model is the best, followed by SVM and neural network. The average RMSE performance of these three models is significantly better than that of decision tree and linear or nonlinear regression model. At the same time, from the stability of RMSE index experimental results and single experimental performance, the performance of random forest is still the best.

Based on the above analysis, it can be seen that the prediction performance of random forest model is the best, followed by SVM and neural network model, and there is a certain gap compared with the other three types of models. Therefore, random forest model should be selected as the optimal prediction model.

IV. CONCLUSION

This paper proposes a method to build a prediction model and accurately predict employee performance based on machine learning algorithm and enterprise human big data. At the same time, in the data construction stage, it focuses on the collection of employee attendance information and data construction. Empirical research has proved the feasibility of accurately predicting employee performance by using machine learning algorithm and employee attendance data.

The contributions are as follows:

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Firstly, a modeling method based on enterprise human big data mining is provided to predict the performance of enterprise employees. The scheme uses enterprise human big data and constructs the optimal employee performance prediction model based on common machine learning algorithms. The empirical research shows that the performance prediction is ideal.

Secondly, a general method of employee attendance information collection and attendance data construction is proposed. The experimental results show that the data set constructed by this method has a good supporting effect on the performance of data mining.

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